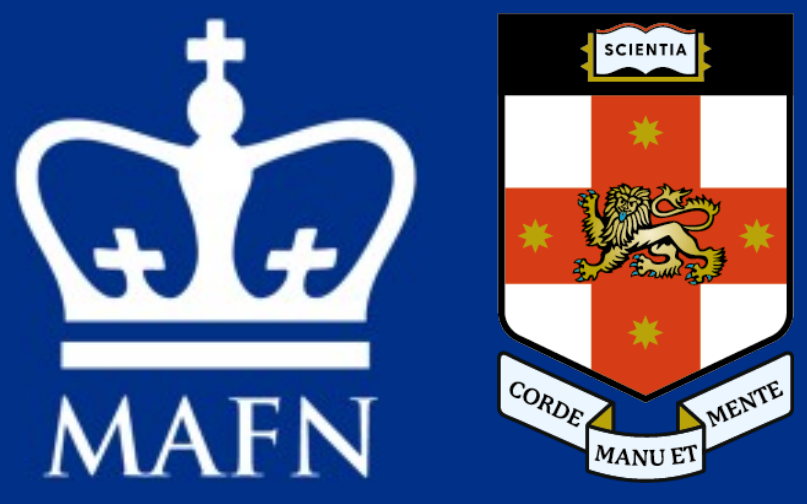




Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

Nigel Li^{1,2}
¹Columbia University, New York

Yutong Han¹
²University of New South Wales, Sydney



Abstract

Reinforcement learning (RL) portfolio allocators are often given a tail-risk limit through a Lagrangian penalty on Conditional Value-at-Risk (CVaR). We show two ways in which this limit can fail with no visible symptom. First, when the penalty and the reward are on different scales, the penalty weight λ never moves away from zero, yet tail metrics still improve because of other parts of the pipeline. Second, the choice of test decides how the learner ranks against classical optimisers. We add a filter that checks each portfolio before it trades and moves it toward minimum variance when its forecast CVaR exceeds the budget. Out of sample, the share of weeks over budget fell from **75.1% to 7.4%**. Across 40 walk-forward folds on two ETF universes, the constraint lowered CVaR₉₉ in **26 of the 27** folds where it was active ($p = 8.8 \times 10^{-6}$).

1 Motivation

In the week of 16 March 2020 the S&P 500 fell about 15% and the VIX closed at 82.69. An RL allocator trained only for return kept its equities. The same allocator trained with a CVaR budget had already moved into Treasuries and cash, and its drawdown was 7.7% against 16.3% (Fig. 1).

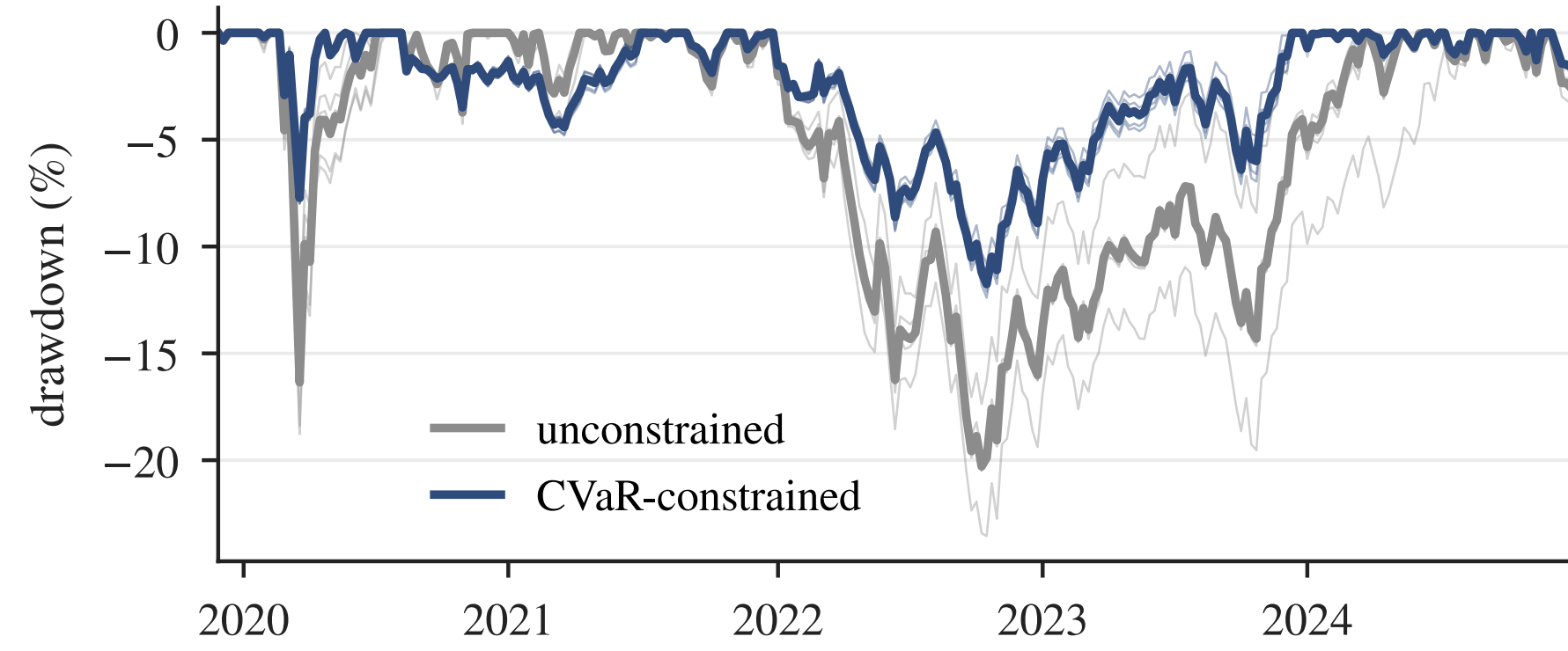


Figure 1. Drawdown over 2020 to 2024. Thick lines are five-seed means and thin lines individual seeds.

A budget of this kind is usually imposed with a Lagrangian penalty whose weight λ grows while the portfolio is over budget [2]. Section 4 asks whether λ actually grows, and Section 5 asks whether the test used to judge the result can be trusted.

2 Tail risk and the budget

CVaR _{α} is the average loss in the worst $(1 - \alpha)$ share of weeks [1]. With $\alpha = 0.95$ it averages the worst 5%. It looks only at losses, and it measures how large they are beyond the Value-at-Risk (VaR) threshold (Fig. 2).

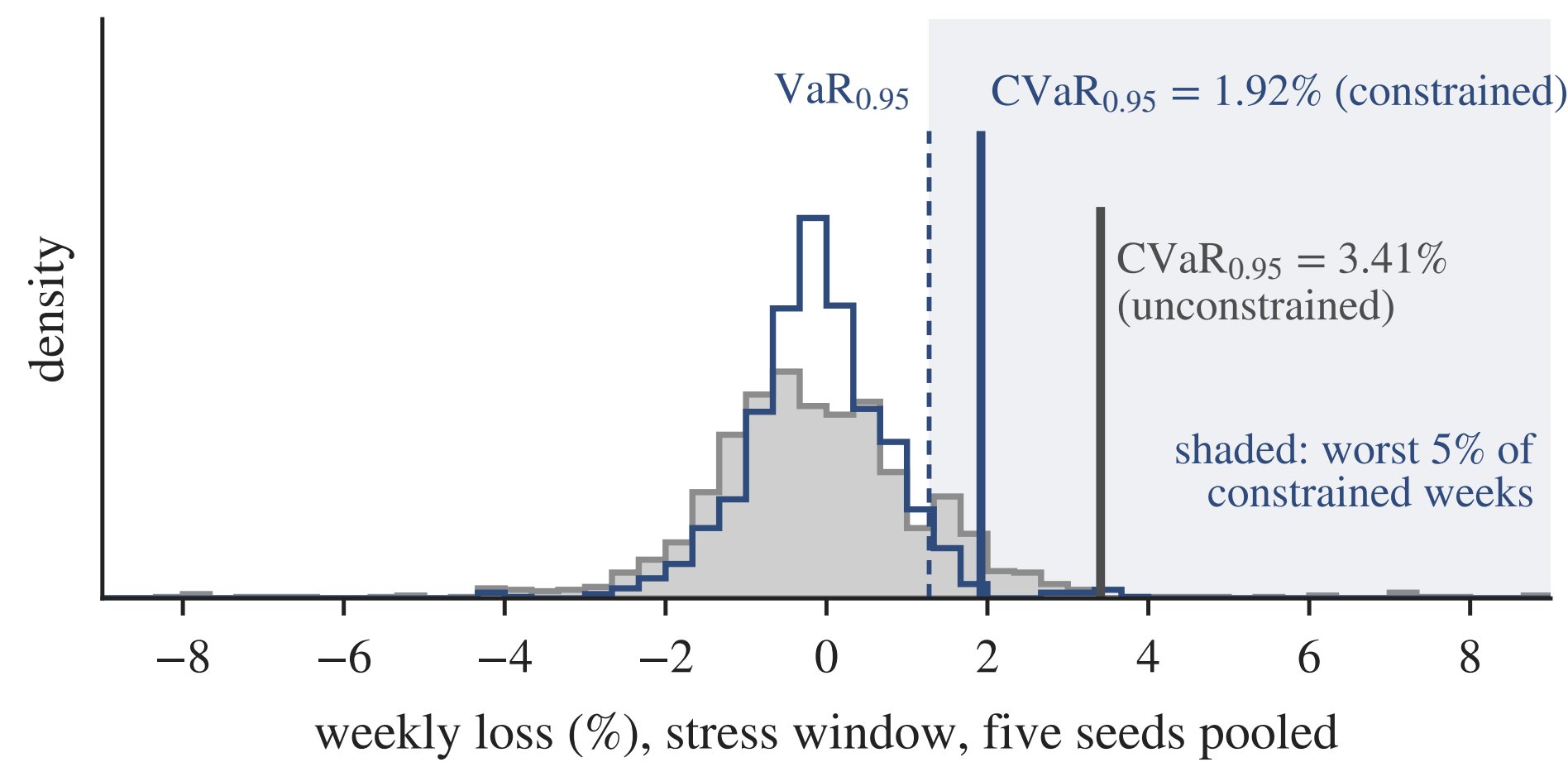


Figure 2. Weekly losses on the 2020 to 2024 stress window, five seeds pooled (1335 weeks per allocator). Each CVaR line is the mean of the losses at or beyond that allocator's VaR (constrained VaR 1.28%, unconstrained 2.00%). Averaged per seed, as in Table 1, the values are 1.89% and 3.33%.

Each week the allocator chooses long-only weights w_t on seven ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL), with at most 40% in one asset, weekly turnover at most 0.50 and a 5 bps cost. The problem is a constrained Markov decision process [2],

$$\max_{\pi} J(\pi) \quad \text{s.t.} \quad \text{CVaR}_{0.95}(L) \leq d = 1.2\%, \quad \text{CVaR}_{\alpha}(L) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[(L - \zeta)^+] \right\}.$$

The data are 887 weekly returns from 2008 to 2024. Models train on 2008 to 2018 and are tested on the 267 weeks that include the 2020 crash and the 2022 selloff. A test week is a breach when the trailing 52-week CVaR_{0.95} exceeds d , and up to 5% of weeks may breach. All measures use weekly net returns $r_1, \dots, r_T$ and wealth $W_t = \prod_{u \leq t} (1 + r_u)$,

$$\text{Sharpe} = \sqrt{52} \bar{r} / s_r, \quad \text{MDD} = \max_t (1 - W_t / \max_{u \leq t} W_u), \quad B = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathbf{1}[\overline{\text{CVaR}}_{0.95}(r_{t-52:t-1}) > d],$$

where $\hat{q}_{1-\alpha}$ is the empirical $(1 - \alpha)$ quantile and B is the breach rate over the test weeks $\mathcal{T}$. Sharpe significance uses the Probabilistic and Deflated Sharpe ratios [4], with per-week $\widehat{SR}$, skewness $\hat{\gamma}_3$, kurtosis $\hat{\gamma}_4$ and N configurations tried,

$$\text{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right), \quad SR_{\text{DSR}}^* = \hat{\sigma}_{SR} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{N_e} \right) \right],$$

where γ is the Euler-Mascheroni constant.

3 Method

The allocator is trained end to end through the portfolio return, and a filter checks every portfolio before it trades (Fig. 3).

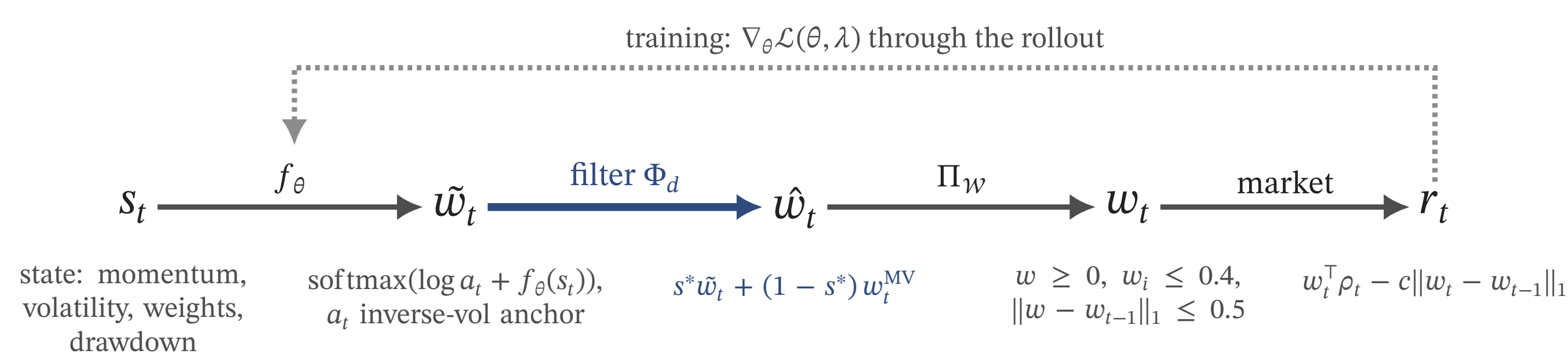


Figure 3. The decision at week t as a composition of maps. The navy map is the decision-time filter.

Training minimises, over random windows of past weeks,

$$\mathcal{L}(\theta, \lambda) = -\text{Sharpe}(r) + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad \lambda \leftarrow \max\{0, \lambda + \eta_{\lambda}(\widehat{\text{CVaR}} - d)\},$$

where $\widehat{\text{CVaR}}$ is the mean of the worst 5% of returns in the window, so its gradient acts only on those weeks. The filter, a safety layer in the sense of [3], forecasts CVaR from the last 104 weeks as $\widehat{\text{CVaR}}(w) = c_{\alpha} \sigma_p(w) - \mu_p(w)$ with $c_{0.95} = 2.06$ (Gaussian). An over-budget proposal is mixed with the trailing minimum-variance portfolio, keeping as much of the proposal as the budget allows,

$$s^* = \max \{s \in [0, 1] : \widehat{\text{CVaR}}(s \tilde{w}_t + (1-s) w_t^{\text{MV}}) \leq d\}.$$

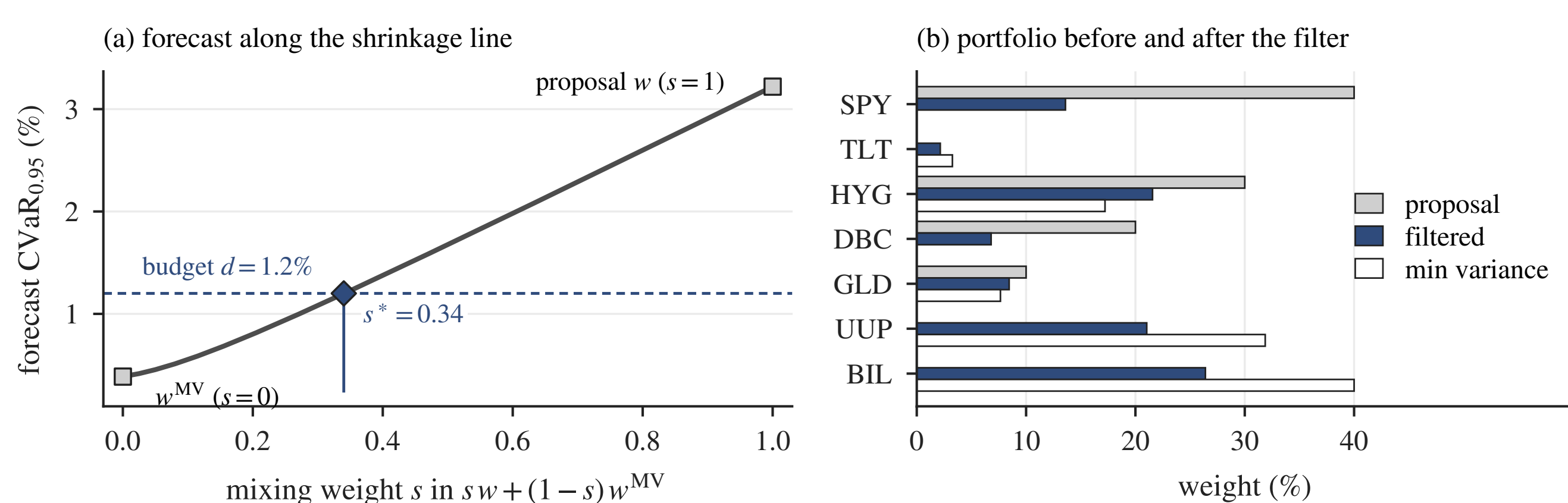


Figure 4. The filter on 16 March 2020 for an illustrative equity-heavy proposal, using the trailing ETF data. (a) Forecast CVaR along the mixing line. The budget is met at $s^* = 0.34$. (b) Weights before and after.

The filter uses only data up to the previous week, and an automated look-ahead test confirms this.

4 Failure 1. The penalty weight never moves

With a dual step $\eta_{\lambda} = 10^{-3}$, λ stayed near zero for all 1500 training updates (Fig. 5). The CVaR excess is about 10^{-2} per week, so each update moved λ by about 10^{-5} . With $\eta_{\lambda} = 5.0$, λ rose to between 0.1 and 0.3. A model-free actor-critic failed in the same way when its cost signal (about 10^{-3}) was left unstandardised next to a unit-scale reward signal.

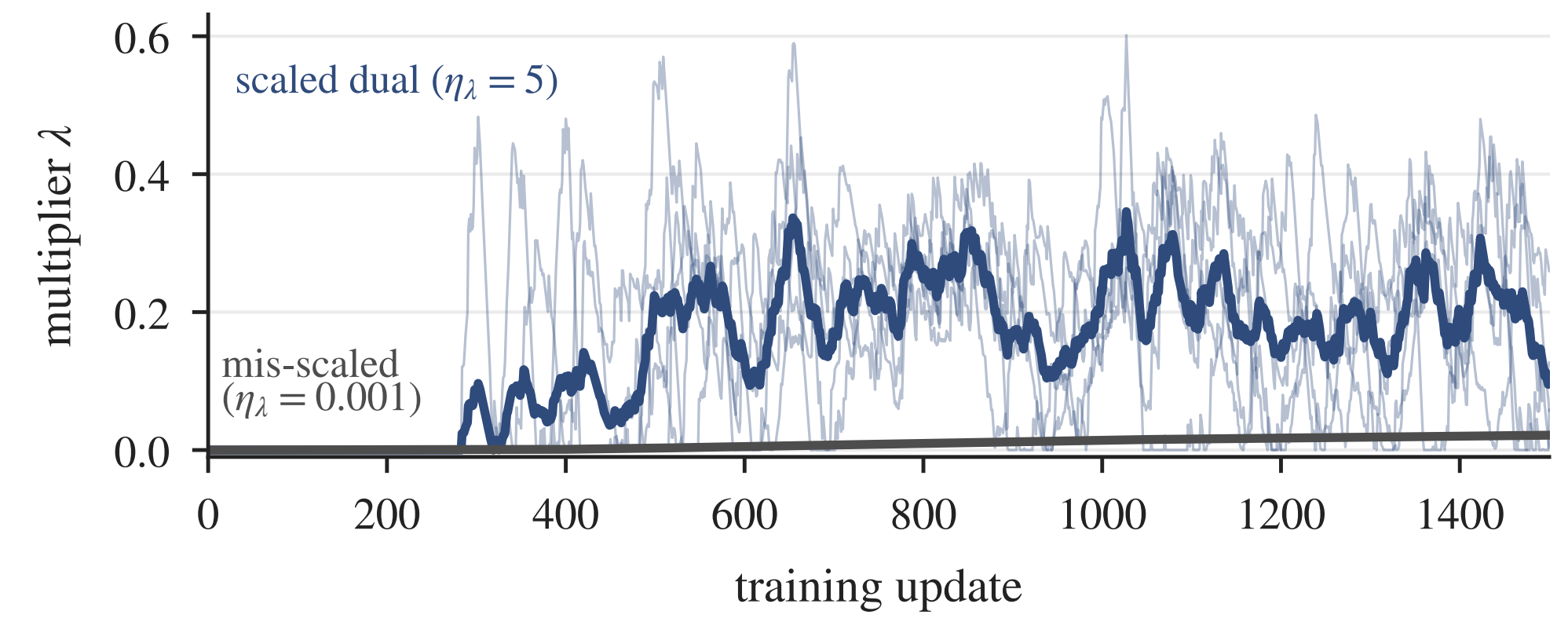


Figure 5. The penalty weight λ during training, five seeds.

Tail metrics do not reveal the failure. Even with λ near zero, the rest of the pipeline (the anchor, the projection and checkpoint selection under the CVaR limit) lowered CVaR₉₉ by 61% against the unconstrained learner (0.025 against 0.065, permutation test $p = 0.0044$).

Plot λ during training. A multiplier that stays at zero means the constraint is inactive.

5 Failure 2. The test decides the ranking

A walk-forward test retrains the model on a rolling window and evaluates it on the next unseen block, repeatedly. On a 31-asset universe, one fixed stress split ranked the constrained learner above minimum variance (Sharpe 0.85 against 0.23). A 20-fold walk-forward reversed the order (0.74 against 1.40, Fig. 6). Ledoit-Wolf shrinkage [5] did not change the split result (0.25). The tail-risk reduction against unconstrained RL held under both tests (Wilcoxon $p = 6 \times 10^{-4}$).

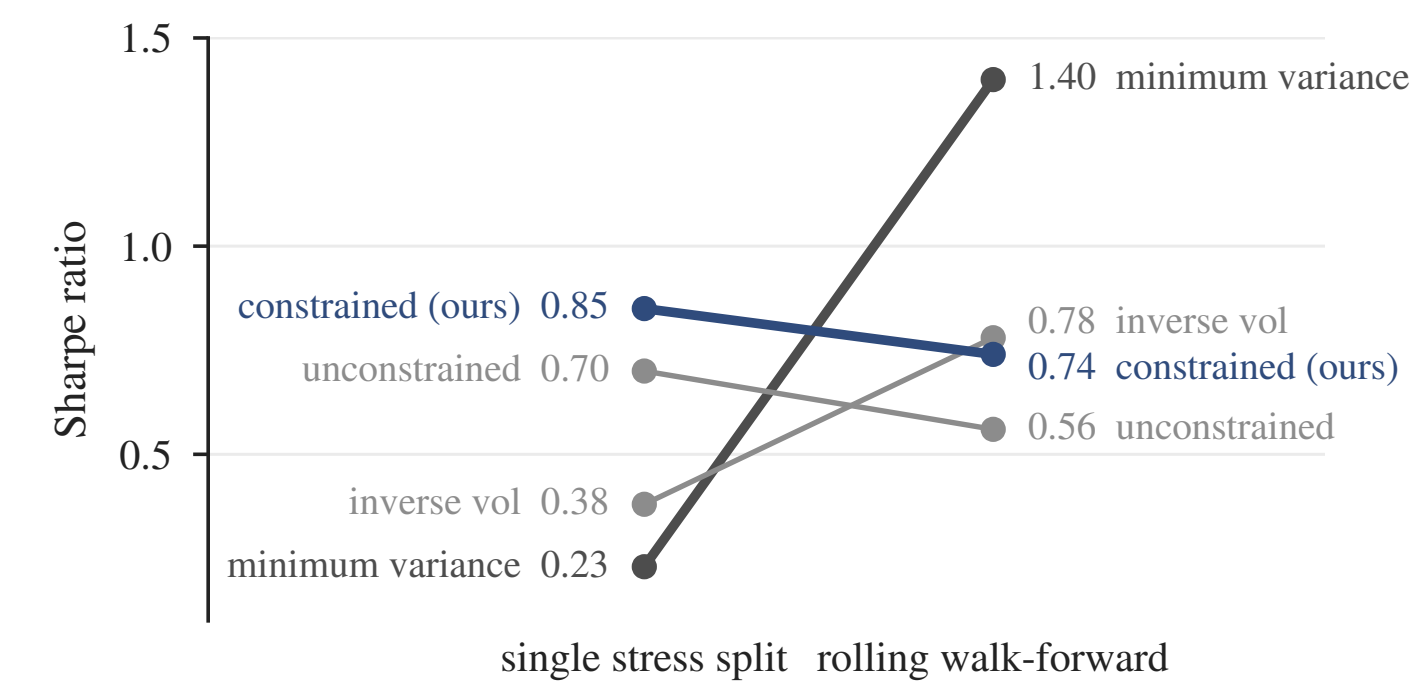


Figure 6. The same four methods ranked by two tests, 31-asset universe.

Compare learners with optimisers on rolling walk-forward folds and report the Deflated Sharpe ratio [4].

6 The filter keeps the portfolio within budget

A larger dual step made λ move, but the portfolio was then over budget in 75.1% of test weeks, although classical rules met the same budget in 92% of weeks. Adding the filter lowered the breach rate to **7.4%**, with CVaR₉₉ of 0.016 and Sharpe 0.851 (Fig. 7). Raising λ during testing without the filter left it at 81.9%.

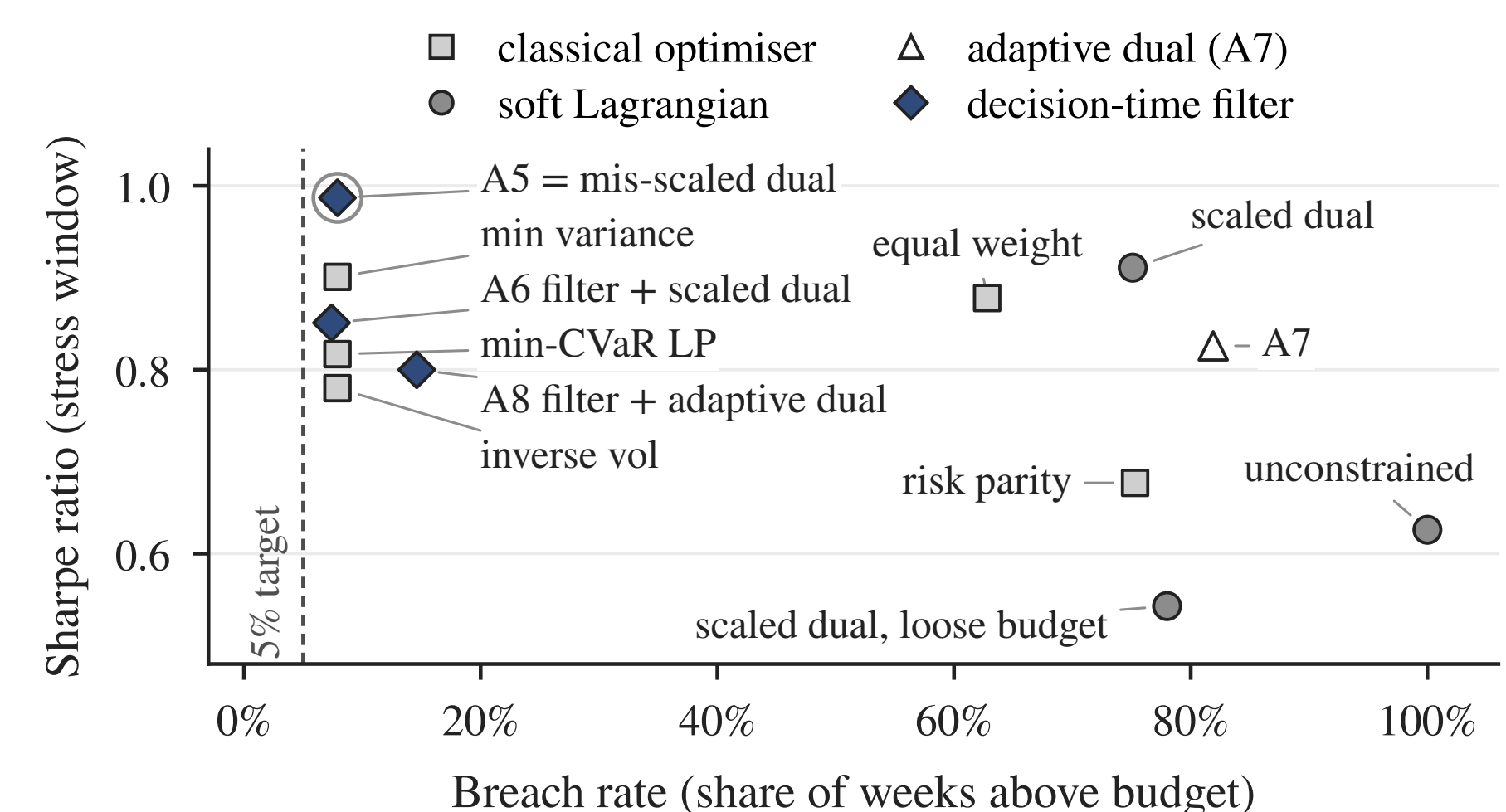


Figure 7. Share of weeks over budget against Sharpe on the stress window. The dashed line is the 5% tolerance.

7 Results across tests

For walk-forward fold k let $\Delta_k = \text{CVaR}_{0.99,k}^{\text{con}} - \text{CVaR}_{0.99,k}^{\text{unc}}$. The Wilcoxon signed-rank test uses $W^+ = \sum_{k: \Delta_k \neq 0} R_k \mathbf{1}[\Delta_k > 0]$, where R_k is the rank of $|\Delta_k|$ among non-zero differences, so tied folds drop out.

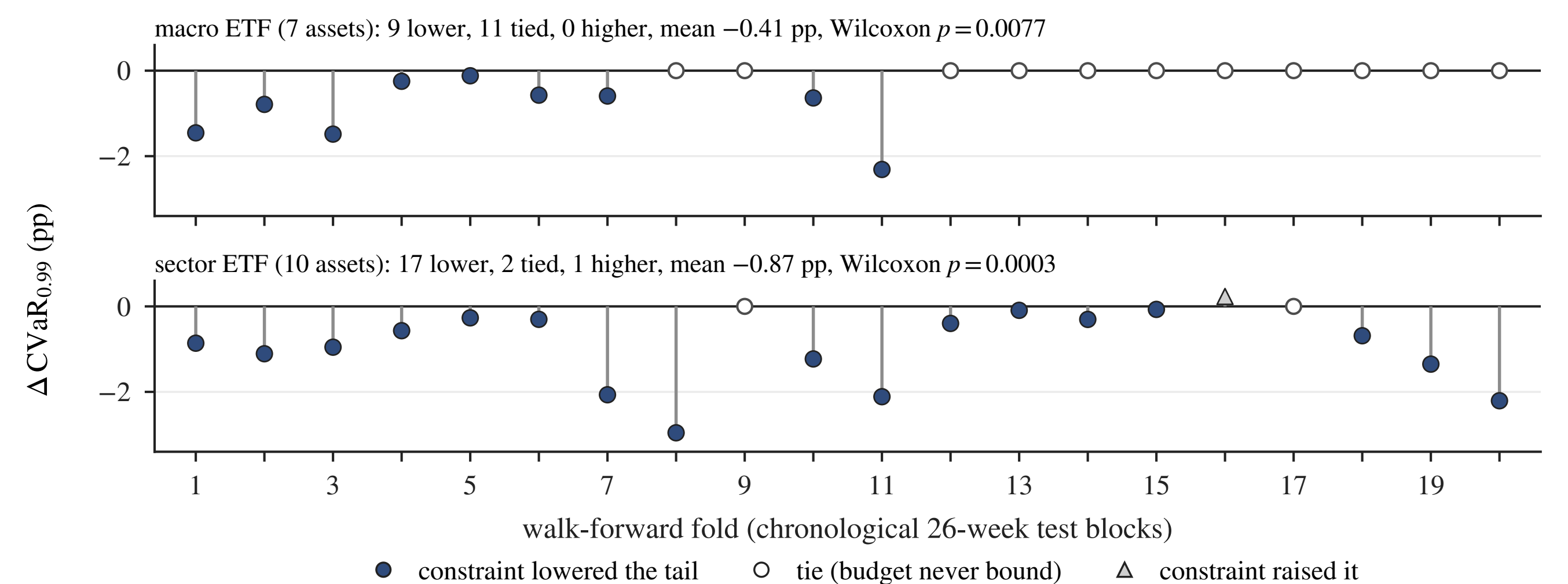


Figure 8. Δ_k for all 40 folds. In a tied fold the budget never binds. Pooled, 26 of 27 untied folds are lower ($p = 8.8 \times 10^{-6}$), and calendar-clustered $p = 3.3 \times 10^{-4}$.

metric	unconstr.	constr.	change
Sharpe	0.63 $\pm$ 0.09	0.88 $\pm$ 0.04	+40%
Sortino	0.61	0.87	+44%
MDD	0.206	0.118	-43%
CVaR ₉₅	0.0333	0.0189	-43%
CVaR ₉₉	0.0647	0.0327	-49%
violations per run	7.2	0.0	-100%

The stress-window reduction in CVaR₉₉ has a 95% BCA interval of $[-0.035, -0.022]$ and replicates on sector ETFs (30% lower at unchanged Sharpe). Both universes pass a Benjamini-Hochberg correction [6]. Against minimum variance the constrained policy did better in high-volatility weeks (Sharpe 2.169 against 1.668) and worse in selloffs (-0.809 against 0.035).

PSR(0) (Section 2) was at least 0.99 on both universes, but the Sharpe advantage over classical baselines did not survive DSR.

8 Conclusion

The constraint lowered tail risk in a way that survived multiple-testing correction, and the filter achieved the out-of-sample compliance that rescaling the dual did not. The learner did not beat rolling minimum variance on Sharpe out of sample. The study uses weekly data with proportional costs, and the Gaussian forecast can understate fat tails.

References. [1] R. T. Rockafellar, S. Uryasev. Optimization of conditional value-at-risk. *J. Risk* 2 (2000). [2] E. Altman. *Constrained Markov Decision Processes*. 1999. [3] G. Dalal et al. Safe exploration in continuous action spaces. arXiv:1801.08757, 2018. [4] D. H. Bailey, M. López de Prado. The deflated Sharpe ratio. *JPM* 40 (2014). [5] O. Ledoit, M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *JMV* 88 (2004). [6] Y. Benjamini, Y. Hochberg. Controlling the false discovery rate. *JRSS B* 57 (1995).

✉ nl2992@columbia.edu nigel.li@unsw.edu.au yh3915@columbia.edu
 ✉ github.com/nl2992/ICAIF_cvar-rl-portfolio-allocator

